



PL-7008

Create agents in Microsoft Copilot Studio

PL-7008 Course schedule

Introduction

1. Get started with Microsoft Copilot Studio
2. Design agent conversations using topics in Microsoft Copilot Studio
3. Build intelligent agents in Microsoft Copilot Studio
4. Add structured automation to agents in Microsoft Copilot Studio

2.

Design agent conversations using topics

Learning Objectives

After completing this module, you will be able to:

1 Create topics in Microsoft Copilot Studio

2 Define the conversation path with nodes

3 Manage topics

4 Work with variables

5 Work with entities

Microsoft Learn module

Design agent conversations using topics

<https://learn.microsoft.com/training/modules/copilot-studio-topics>



Create and configure agent topics

Adding topics in Copilot Studio



From blank: Create topics by adding trigger phrases

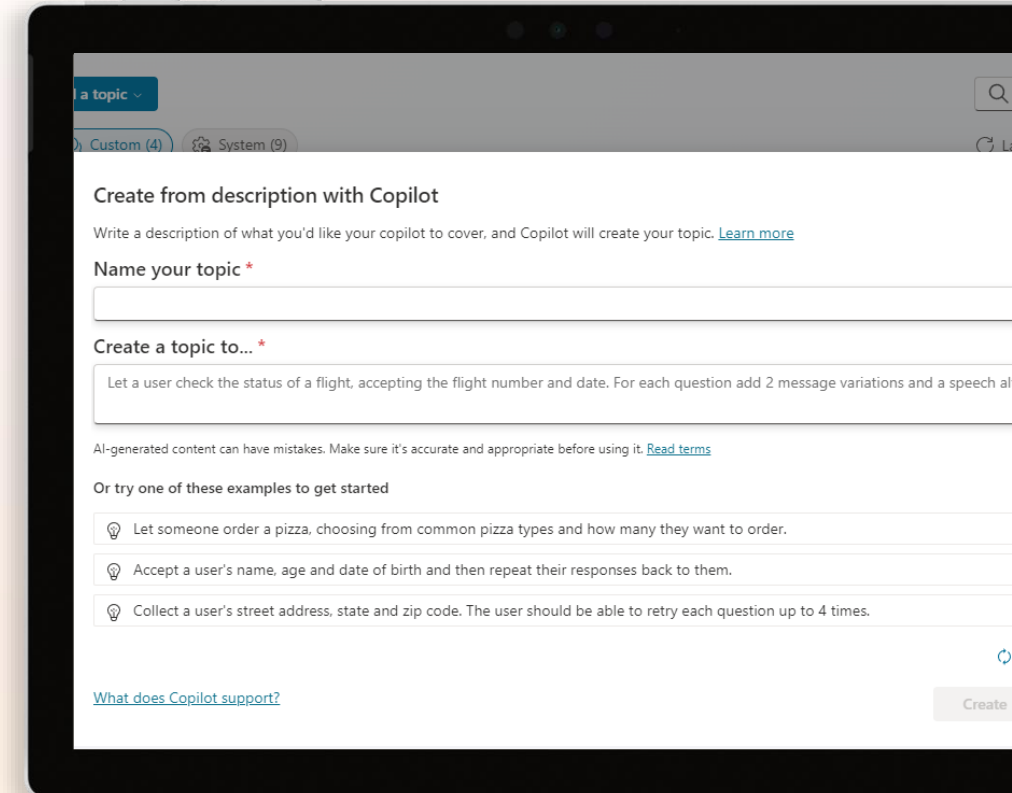


From a description using Copilot: Author topics using natural language

Author topics using natural language

Create from description

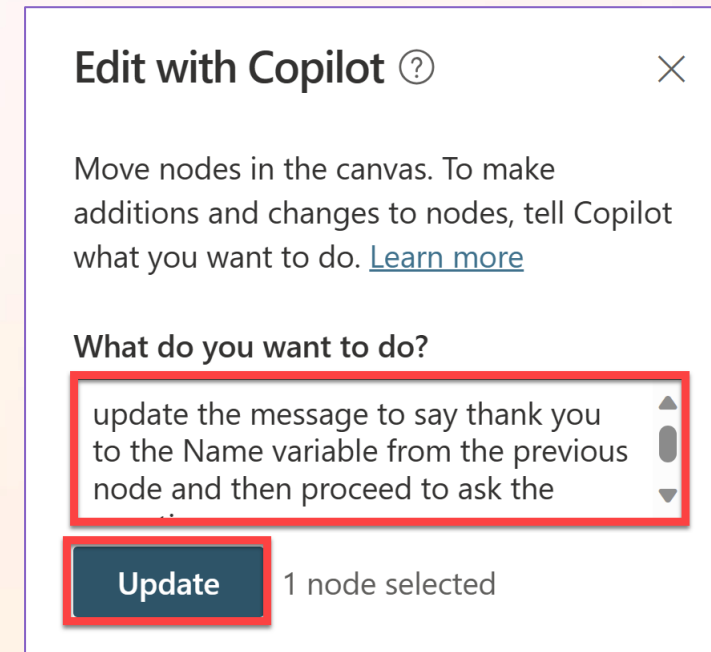
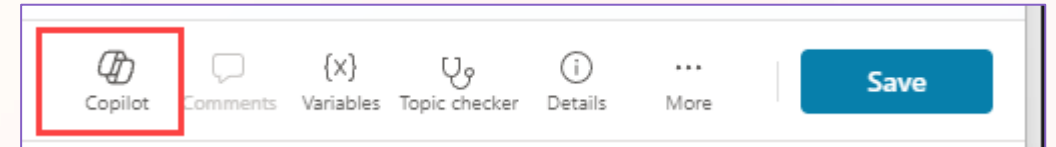
- Describe the topic in natural language
- Topic is created with conversational responses and multiple types of nodes



Update nodes with natural language

Change topic

- Select Copilot icon
- Select a node to change a node
- Select canvas to add a node
- Edit with Copilot pane
- Describe what you would like to change
- Use simple, plain language to direct the AI



Demo

Example: Topic created with Copilot

Real Estate Booking Service

OverviewKnowledgeTopicsActionsAnalyticsChannelsPublished 7/14/2024

PublishSettings

Test

Customer Details

CopilotCommentsVariablesTopic checkerDetailsMore

Save

↶ | ✂ | 📄 | 🗑

Trigger

Phrases

Phrases

customer details

get customer information

retrieve customer details

customer profile

view customer data

+

Question

Text

What is your full name?

Identify

Person name

Save user response as

FullNamestring

+

Question

Text

+

✖

🔍

👁

🔍

🕒

🔍

👤

Edit with Copilot

Move nodes in the canvas. To make additions and changes to nodes, tell Copilot what you want to do. [Learn more](#)

What do you want to do?

Explain what you want to change or add from the topic. Select a node to be more specific.

UpdateNo nodes selected

What you asked for

"Ask the customer for their full name and email address"

What we added

Question

What is your full name?

Question

What is your email address?

Examples that work

Add a question to ask the user for their name.

Summarize the information gathered from the user in an Adaptive Card.

Add a question to ask the user to choose their preferred contact method, choosing from email, telephone or SMS.

Topic details

Generative AI orchestration

- Name
- Description
- Model display name
- Model description

The screenshot displays the 'Topic details' interface. At the top, a navigation bar includes icons for Copilot, Comments, Variables, Topic checker, and a highlighted 'Details' tab. The main workspace shows a 'Trigger' card with a lightbulb icon and a description: 'The agent chooses' followed by an 'Edit' link. Below this, it says 'Describe what the topic does' and 'Books a real estate showing for a customer'. A plus sign in a circle is at the bottom of the card. On the right, a 'Topic details' sidebar is open, showing fields for 'Name' (Book Showing), 'Description' (Enter a description), 'Model display name' (Book Showing), and 'Model description' (Books a real estate customer). The 'Details' tab in the top navigation bar is highlighted with a red box.

Trigger phrases

Classic orchestration

- Manually add trigger phrases

The screenshot displays the Microsoft Bot Framework Classic orchestration interface. On the left, a 'Trigger' node is highlighted with a blue border. Inside the node, there is a card titled 'User says a phrase' with a red box around the 'Edit' button. Below the card, the text 'Phrases' and 'No phrases to show' is visible. A plus sign icon is located below the node. On the right, a configuration panel for the 'Phrases' property is shown, enclosed in a red border. The panel contains the following text: 'Phrases', 'To start learning, the agent needs 5-10 short trigger phrases.', a link 'Show writing tips', a section 'Add phrases', a paragraph 'Phrases teach the agent different ways someone might ask about this topic. Natural language understanding helps identify a topic based on meaning and not exact words.', a text input field with the placeholder 'Enter text' and a plus icon, a paragraph 'To add items in bulk, paste in line-separated items or use Shift+Enter to create line separation or', a link 'upload a file | download', and a section '> Advanced'.

Add conversation nodes to topics

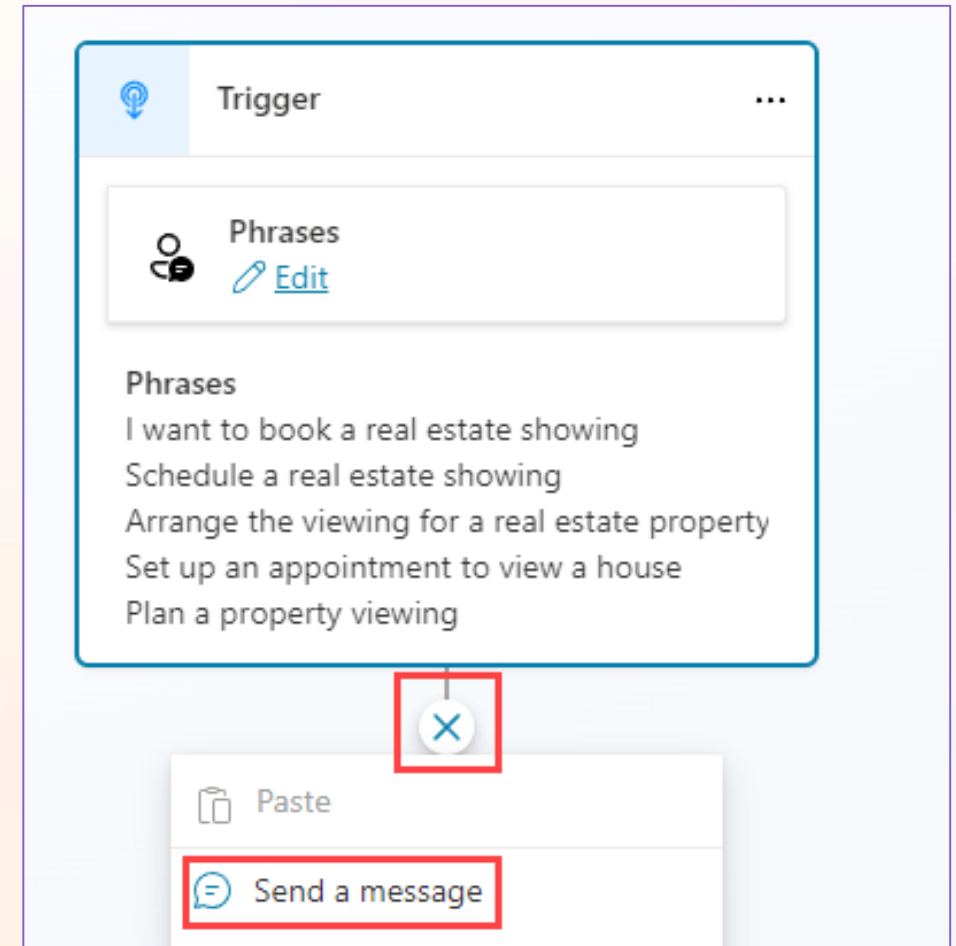
Node types

- 1 Send a message:** Displays a message to the user, with optional formatting, images, video, cards, or quick replies.
- 2 Ask a question / Ask with adaptive card:** Captures information from the user and saves it to a variable, either as a question or an adaptive card.
- 3 Add a condition / Variable management:** Branches the topic flow, and sets, parses, or clears variable values.
- 4 Topic management:** Redirects to another topic, transfers to a live agent, or ends the topic or conversation.
- 5 Add a tool / Add an agent / Advanced:** Calls a workflow or tool, calls another agent, or uses generative answers, HTTP requests, and events.

Conversation nodes

Add node

- Conversation nodes represent customer interactions or actions that can be inserted into a topic's conversation path
- Click on the + icon to add further nodes



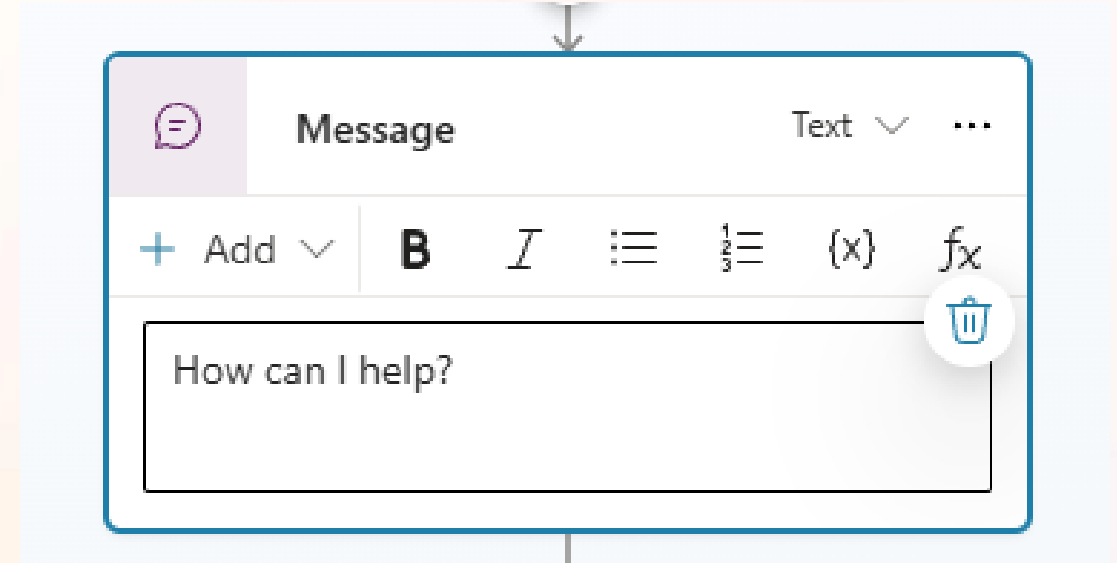
Message node

Message properties

In the text box, enter the message you want the agent to send

Use the tools in the node's menu bar to:

- Apply basic formatting (bold, italic, bulleted, numbered)
- Insert variables
- Add message variations
- Insert an image
- Insert a video
- Add a basic card
- Add an Adaptive Card
- Add quick replies



Question node

Question properties

- **Ask a question:** The question text that you want to present to the user.
- **Identify:** Defines what the agent should be listening for in the user's response.
 - User's entire response
 - Entity
- **Save response as:** Variable

The image shows the configuration interface for a 'Question' node in a chatbot builder. The main configuration area on the left includes a text input for the question, an 'Identify' dropdown set to 'User's entire response', and a 'Save user response as' dropdown set to 'PropertyName' with a type of 'string'. The 'Variable properties' sidebar on the right shows the 'Variable name' as 'PropertyName', the 'Type' as 'string', and the 'Reference' as the current question node. The 'Usage' section has three options: 'Topic (limited scope)' (selected), 'Receive values from other topics', 'Return values to original topics', and 'Global (any topic can access)'.

Question node

Question behavior

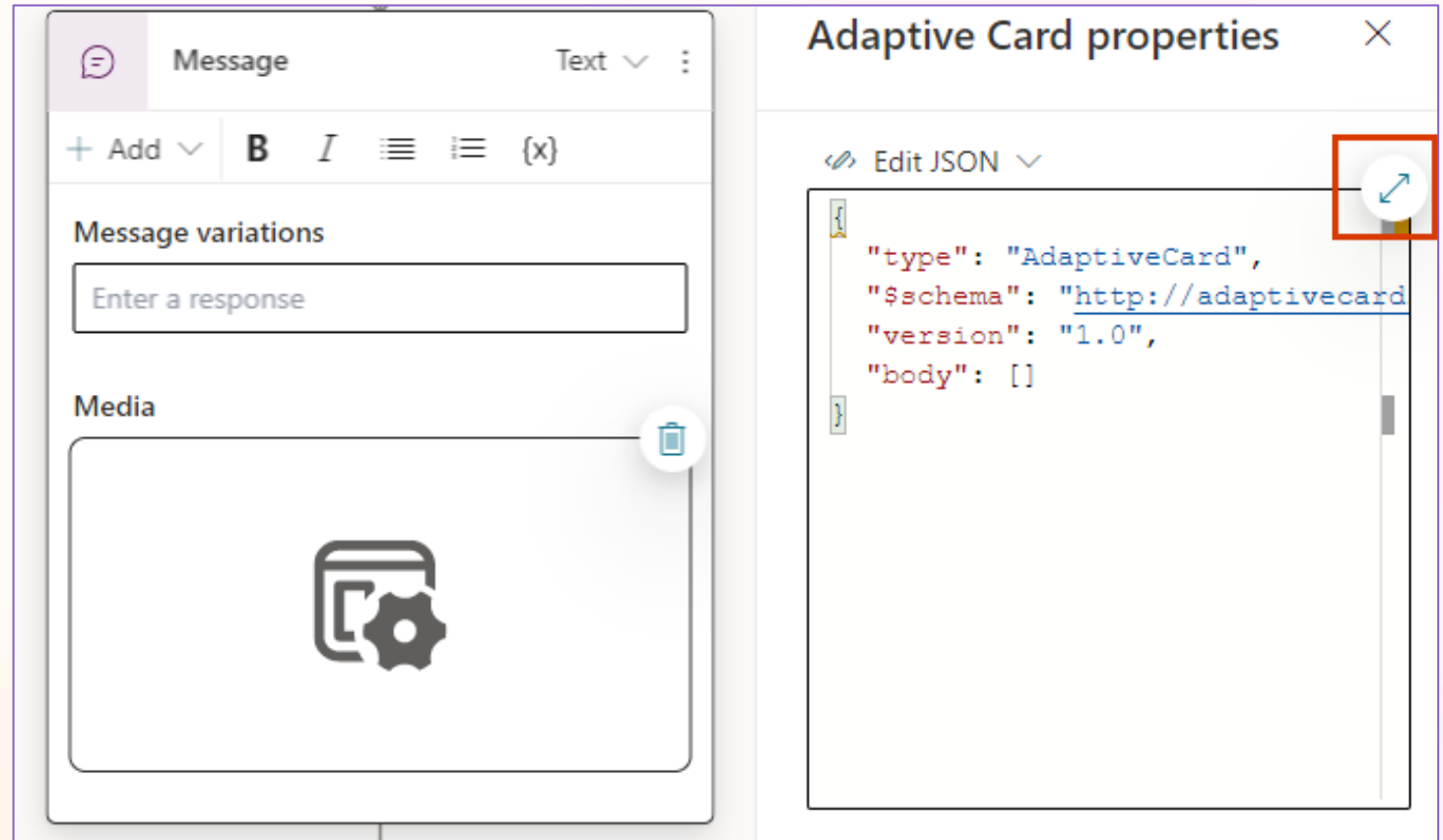
- Skip behavior
- Reprompt
- Additional entity validation
- No valid entity found
- Interruptions

The screenshot shows the configuration interface for a 'Question' node. On the left, a card titled 'Question' is displayed with a 'Text' dropdown menu. A red box highlights the three-dot menu icon next to the 'Text' dropdown. Below the card, the text 'Booking date?' is visible, followed by an 'Identify' section with a 'Date and time' entity type selected. The 'Save response as' section shows a variable 'bookingDate' of type 'datetime'. On the right, a 'Question properties' panel is open, and a red box highlights the 'Question behavior' section, which contains the text: 'Control how your bot asks this question and identifies the entity.'

Adaptive Card node

Adaptive Card Properties

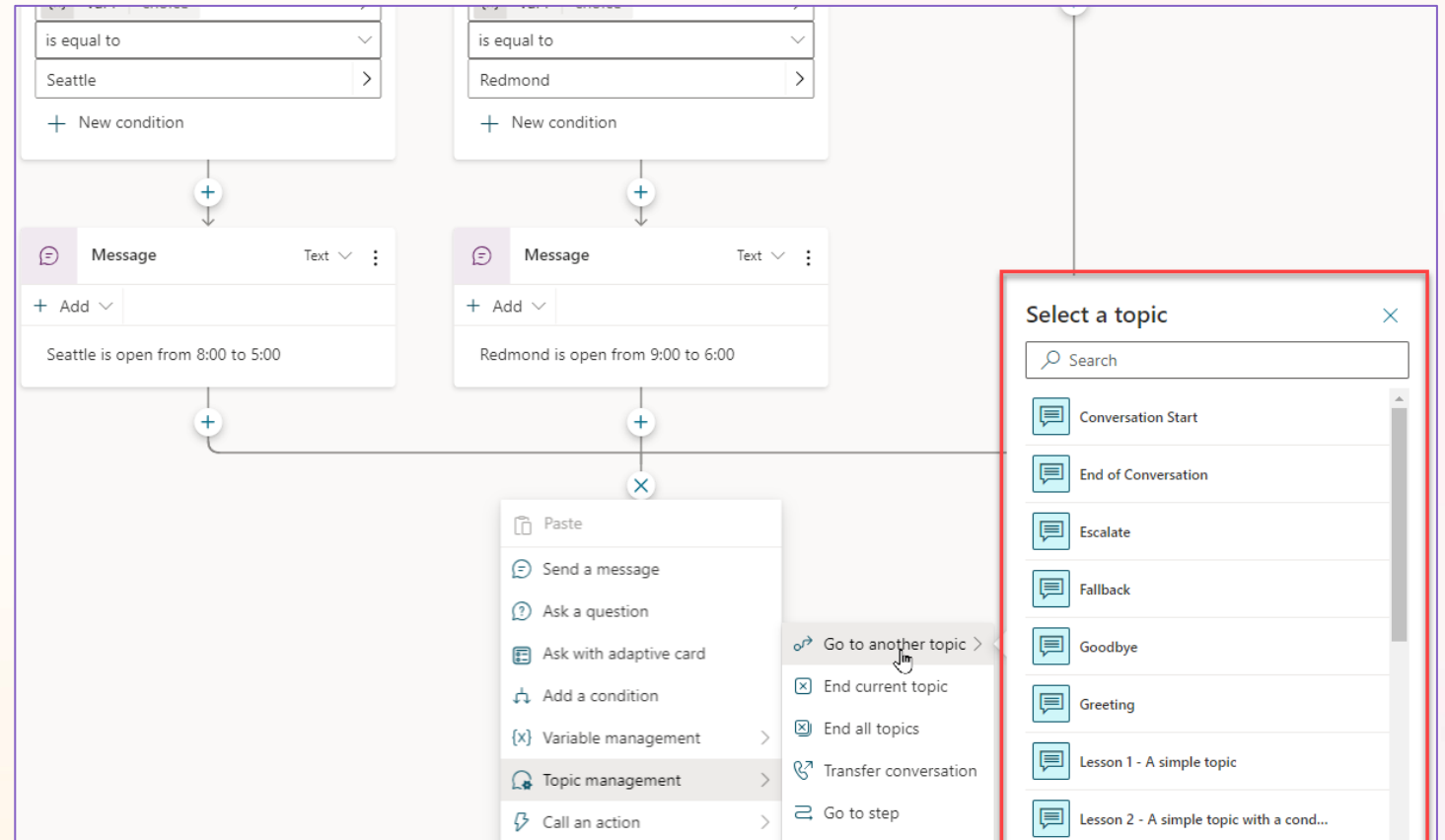
- Define JSON for the card
- JSON is transformed into native UI that automatically adapts to its surroundings



Topic management node

Controls flow between topics

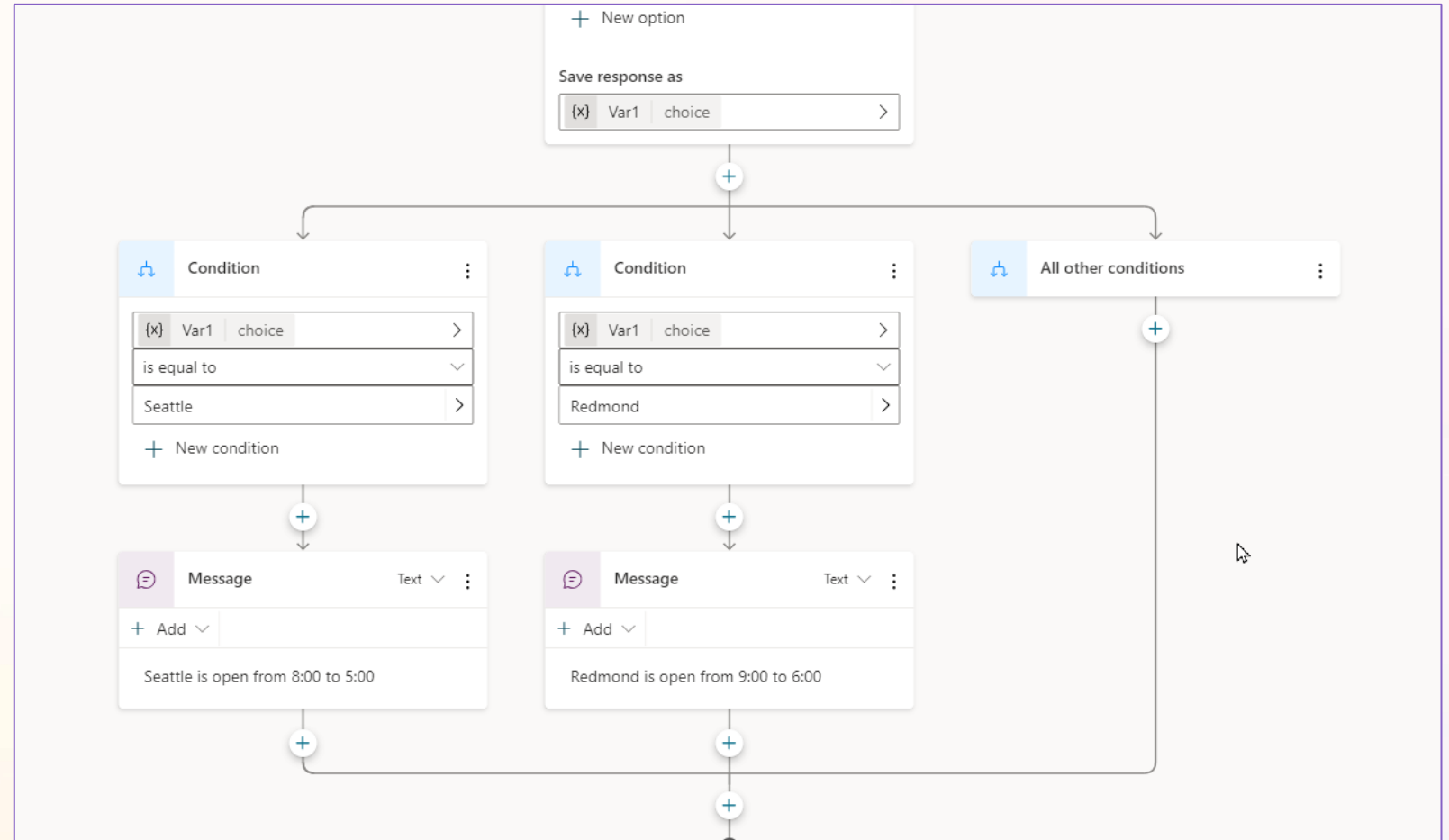
- Go to another topic
- Go to step
- End current topic
- End all topics
- Transfer conversation
- End conversation



Condition node

Branching

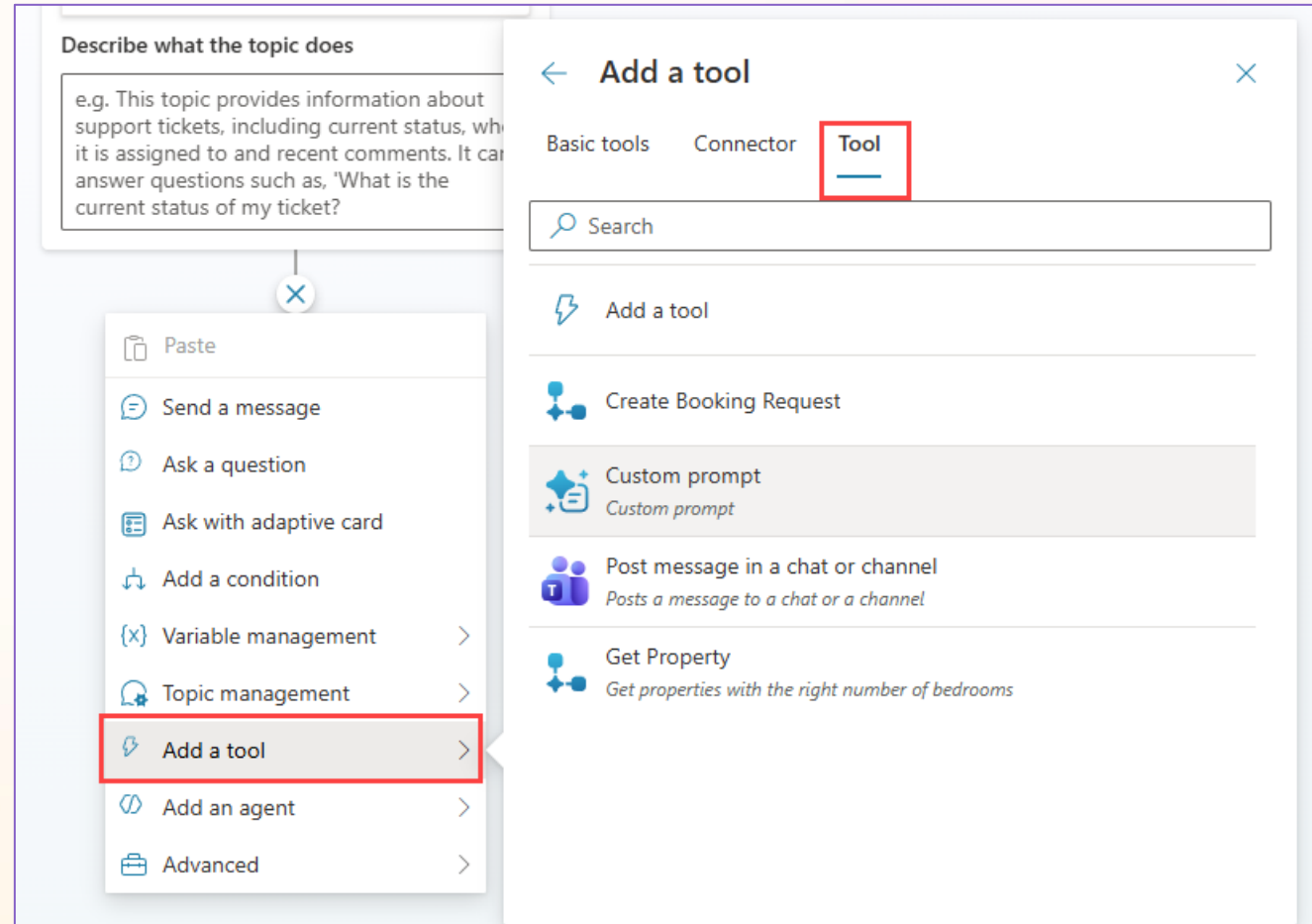
- Compare variable to a value



Add a tool node

Tabs

- Basic tool
- Connector
- Tool



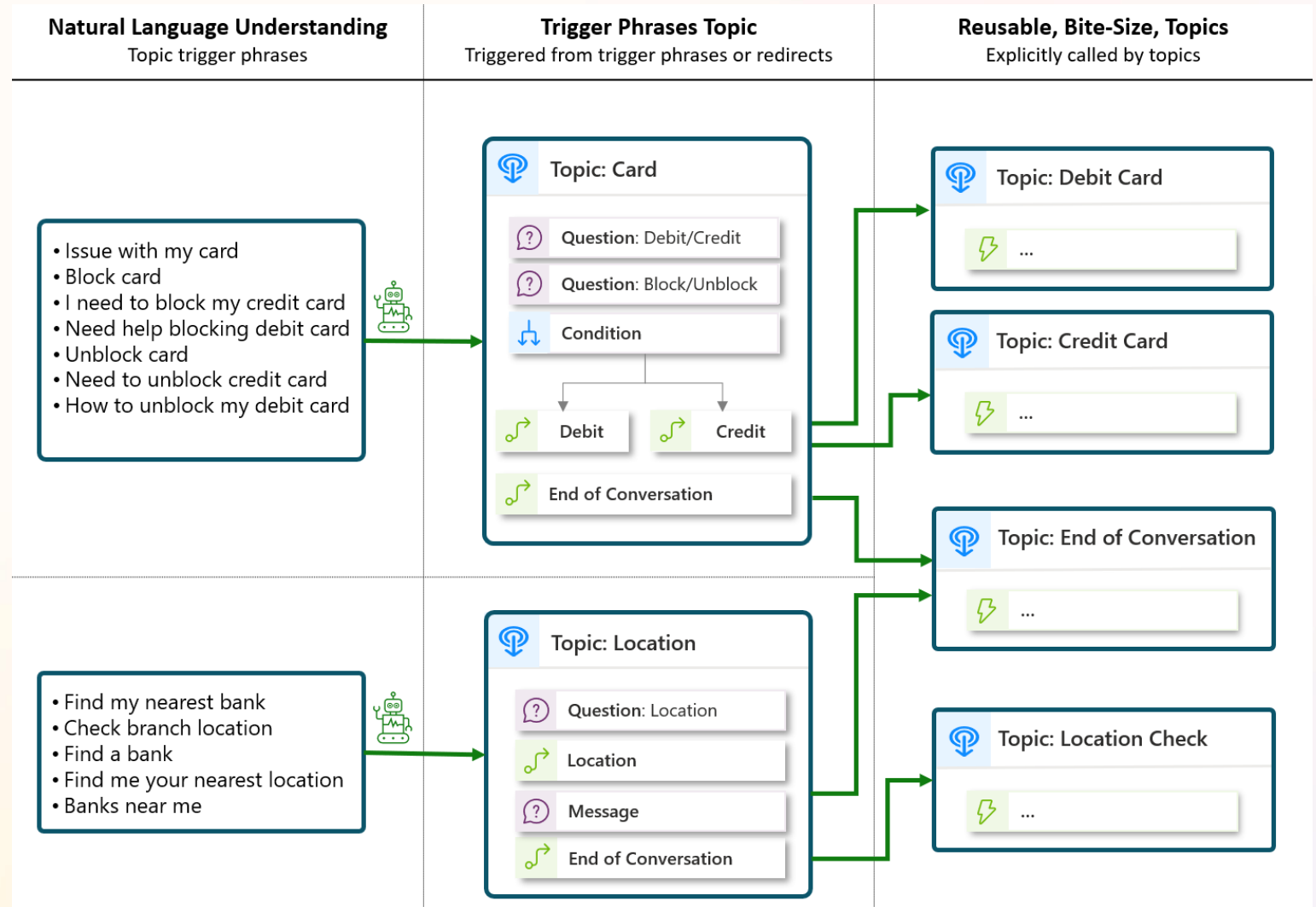
Demo – Add nodes

Manage topics

Authoring in Copilot Studio

Topic best practices

- Create and organize topics in a way that's manageable and maintainable for your context
- Create bite-size topics
- Create reusable topics
- Avoid topic overlap
- Use entities to reduce the number of topics



Manage topics

Delete topics

- Remove topics not required

Name	Trigger
 Goodbye	 Phrases
 Greeting	 Phrases
 Lesson 1 - A simple topic	 Phrases
 Lesson 2 - A simple topic with a cor	 Details  Make a Copy  Delete
 Lesson 3 - A topic with a condition,	
 Start Over	
 Thank you	 Phrases

Disable topics

- Turn off topics not required

Build copilots  Overview  Knowledge  Topics  Actions  Analytics  Channels						
 Add a topic All Custom (7) System (9) Search custom topics Last refreshed now						
Name	Trigger	Description	Editing	Last modified	Errors	Enabled
 Goodbye	 Phrases	This topic triggers ...		MOD Administrat...		 On
 Greeting	 Phrases	This topic is trigger...		MOD Administrat...		 On
 Lesson 1 - A simple topic	 Phrases			MOD Administrat...		 On
 Lesson 2 - A simple topic with a condi...	 Phrases			MOD Administrat...		 On
 Lesson 3 - A topic with a condition, va...	 Phrases			MOD Administrat...		 On
 Start Over	 Phrases			MOD Administrat...		 On
 Thank you	 Phrases	This topic triggers ...		MOD Administrat...		 On

Work with entities

Entities

City

Description

City names, extracted as a string

Method

Prebuilt

Usage

Defines the extracted text for entity.

User input	Entity	Saved value
The business headquarters is in Seattle	Seattle	Seattle
I love NYC	NYC	New York City
I'm going to Vancouver B.C.	Vancouver B.C.	Vancouver B.C.

Show example of user input.

Shows the saved value stored in variable.

Using an entity

Question node

- Select an entity for Identify
- Agent smartly recognizes an entity and extracts the relevant information from the user input and save it to a variable



Question

Text ☐ 

What city do you live in?

Identify

☐ City 

Save response as

 {x}

UserCity

string



Prebuilt entities

Commonly used information

- Person
- Location
- Numbers
- Date and times
- Colors

Settings

Generative AI

Agent details

Security

Connection Settings

Authoring Canvas

Entities

Skills

Voice

Languages

Language understanding

Component collections

Advanced

+ Add an entity

Name	Description	Method
Age	Age of a person, place, or thing, extracted as a number	Prebuilt
Boolean	Positive or negative responses, extracted as a Boolean	Prebuilt
City	City names, extracted as a string	Prebuilt
Color	Primary colors and hues on the color spectrum, extracted as a string	Prebuilt
Continent	Continent names, extracted as a string	Prebuilt
Country or region	Country and region names, extracted as a string	Prebuilt
Date	Dates, days of the week, and months relative to a point in time, extracted as a string	Prebuilt
Date and time	Dates, times, days of the week, and months relative to a point in time, extracted as a string	Prebuilt
Date and time without timezone	Dates, times, days of the week, and months relative to a point in time, extracted as a string without the timez...	Prebuilt
Duration	Lengths of time, extracted as a string, in standard TimeSpan format	Prebuilt

Closed list entity

List of items

- Add names
- Add synonyms

Outdoor Store Categories

Name *

Outdoor Store Categories

Description

Description (optional)

Method
List

The bot will try to match an item on the list based on what the customer says.

Modified by
2 minutes ago

Smart matching
☒ on

The Smart matching option enables the bot's understanding of natural language. This can help match misspellings, grammar variations, and words with similar meanings.

If the bot isn't matching enough related words, enhance the bot's understanding

List items

Enter item

Add

Item	Synonyms
Baseball	+ Synonyms
Basketball	+ Synonyms
Biking	+ Synonyms
Bowling	+ Synonyms
Camping	+ Synonyms
Hiking	+ Synonyms
Cricket	+ Synonyms

Save

Close

These items will be used as values stored in variables.

Using a closed list entity

Question node

- Choose a closed list entity for Identify
- Must select the list values from entity

What type of property do you want to...

Identify

Property Type >

Select options for user

Options for user

Apartment

House

Condominium

Duplex

User options for Property Type

Search

List values (4) Display

Apartment	✓
House	✓
Condominium	✓
Duplex	✓

Create a custom entity

Entity types

- Closed list
- Regular expression

Create an entity



Make your copilot smarter with entities.

An entity is a piece of information (person, place, thing, or concept) that the copilot might want to pick out from what your customer says.

[Learn how to use entities](#)

Choose a method

Closed list

Define a list. Use when you have an exact list of items that is a manageable size (small to medium).

Example: List of sizes

- Extra small
- Small
- Medium
- Large

Regular expression (Regex)

Define a specific logical pattern, such as an ID number, credit card number, IP address, and more.

Example: Article ID pattern that starts with "kb" and ends with 6 digits, such as 'kb123456'.

`kb[0-9]{6}`

Regular expression entity

Regex

- Specify a regex pattern
- .NET regular expressions syntax
- Use cases
 - Find specific character patterns
 - Validate text to ensure that it matches a predefined pattern
 - Extract, edit, replace, or delete text substrings

Regex selection

Name *

Regex selection

Description

This is a regex to match on user selections.

Pattern *

Enter regular expression pattern

The pattern of a regex type cannot be empty

Method

Regular expression (Regex)
The copilot will extract any text that matches the specific pattern.
[Learn how to use entities](#)

Examples

Knowledge base article ID that starts with "kb" and ends with 6 digits, "kb123456" for example.
kb[0-9]{6}

Credit card number that has 13 or 16 digits and starts with a 4.
4[0-9]{12}(:[0-9]{3})?

Tracking number that starts with 9 digits and ends with one letter (upper or lower case), "100456789A" for example.
(?i)\d{9}[A-Z]{1}

Save

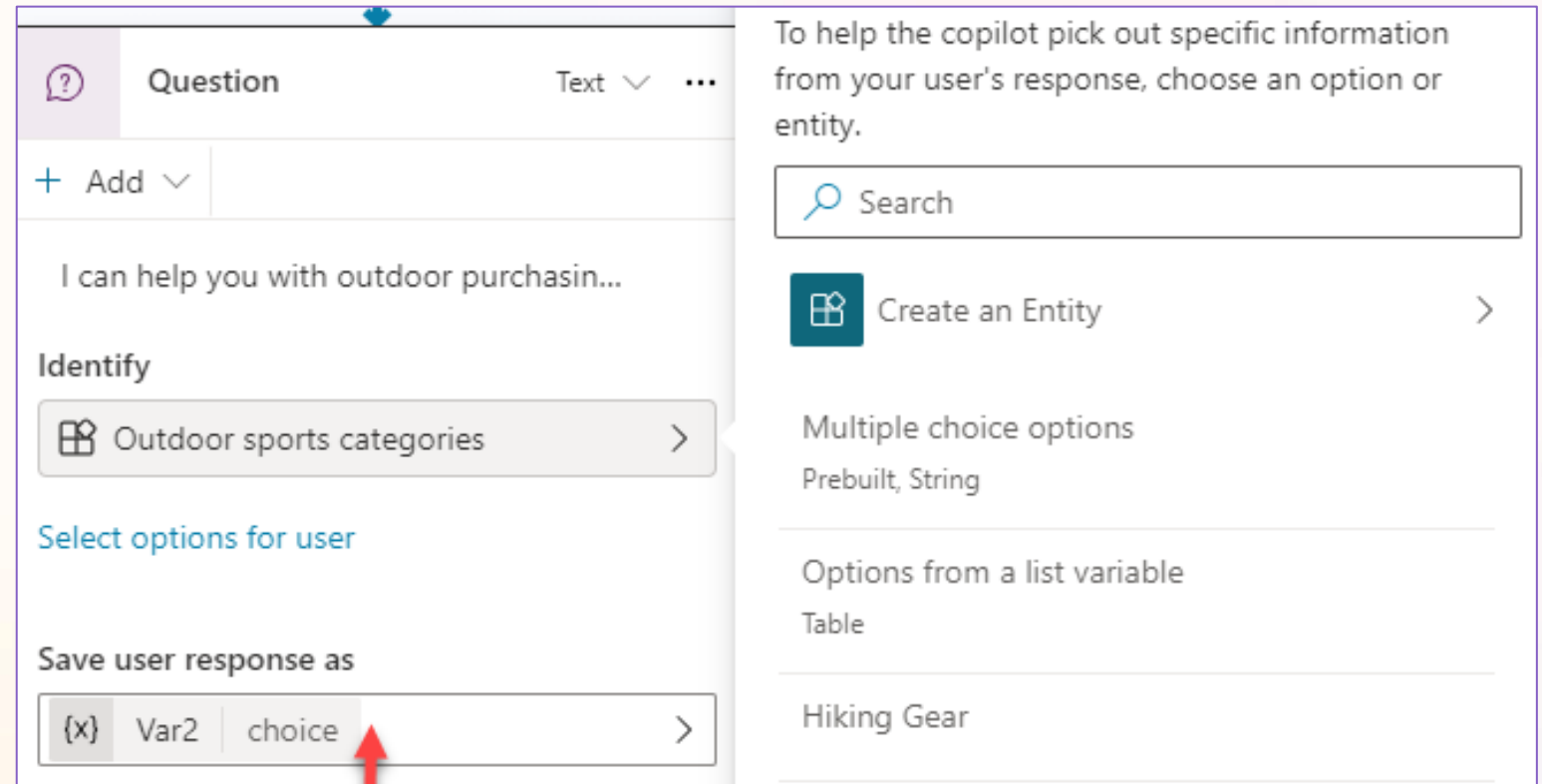
Close

Work with variables in your Copilot Studio agent

Create variables

Create in Question node

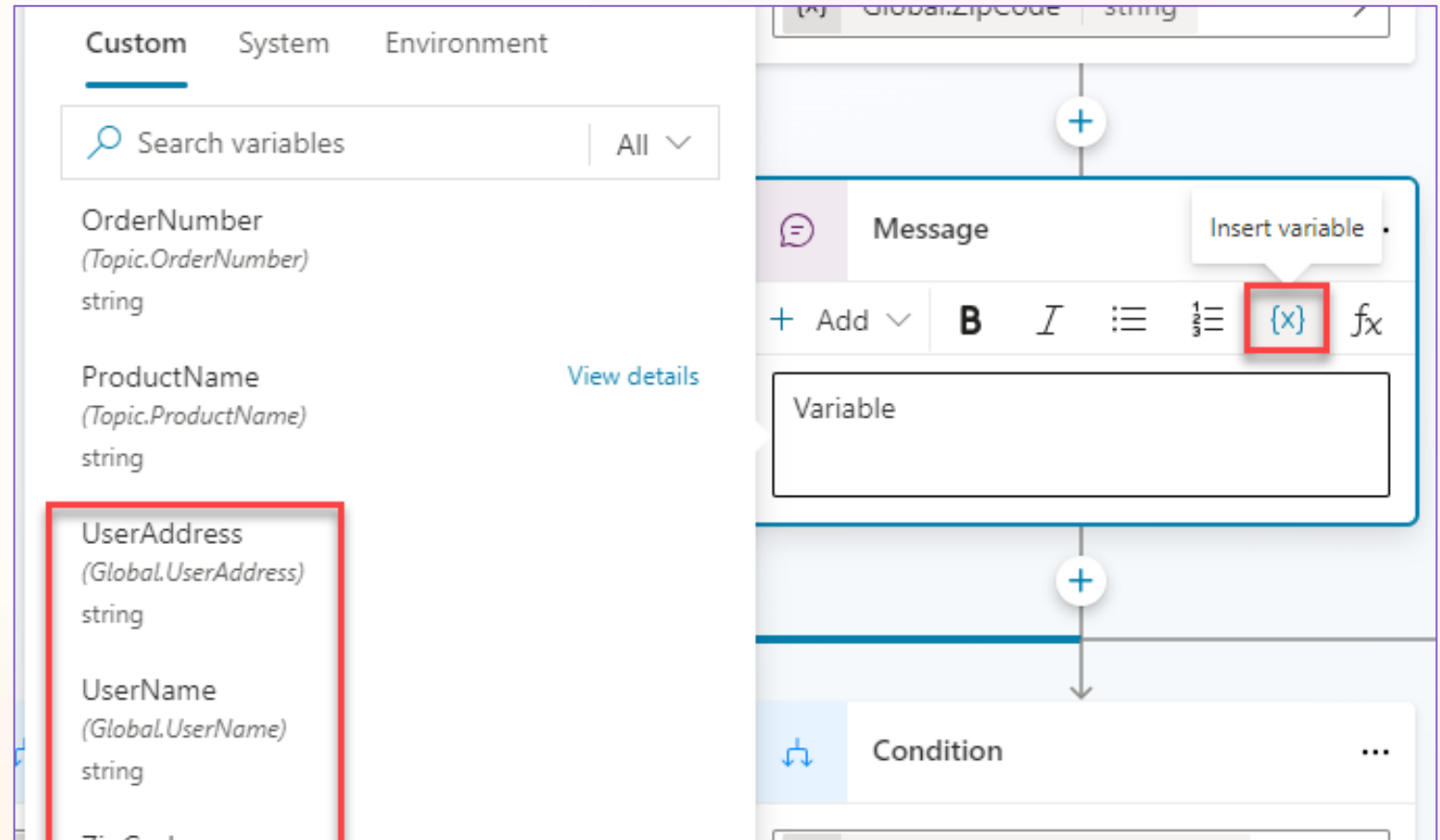
- Created automatically
- Rename



Use variables

{x} icon

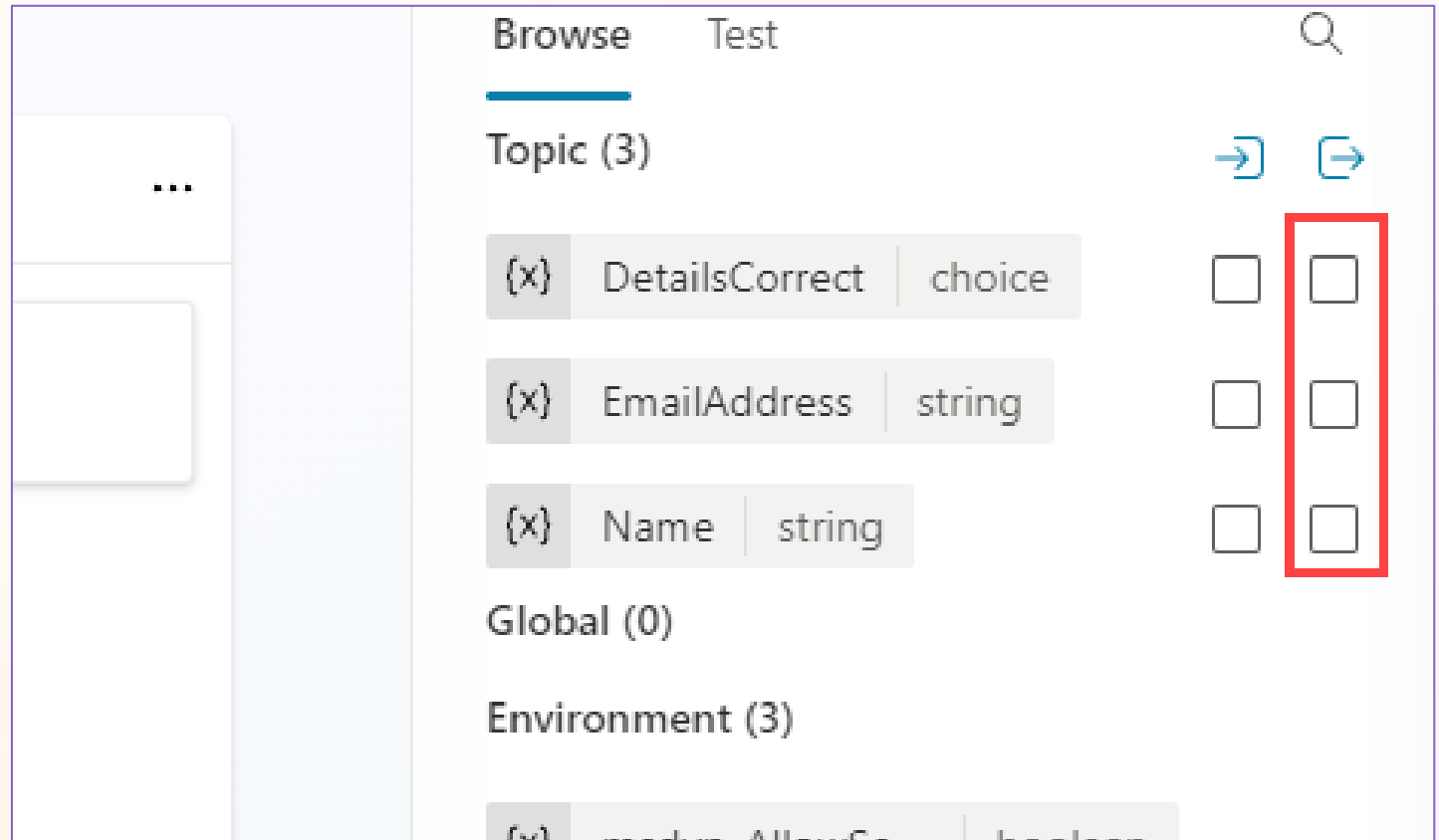
- Insert variable into a node



Pass variables between topics

Variable properties

- Receive values from other topics
- Return values to original topics



Variable scope

Variable types

- Topic (limited scope)
- Global
- System
- Environment

Variable properties

×

Variable name

DetailsCorrect

Type

choice

Reference

 Question
Are the details correct?

⋮

[View all references](#)

Usage

☒ Topic (limited scope)

☐ Receive values from other topics

☒ Return values to original topics

☐ Global (any topic can access)

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Global Variable

Variable names

- Variable name has a prefix string of **Global** to differentiate it from topic-level variables

Trigger ...

The agent chooses [Edit](#)

What the topic does

es customer details

+

Question Text ...

get your name?

Person name >

er response as

Global.PersonName string >

+

Variable properties

Variable name

Global. PersonName

Type

string

Reference

Question
Can I get your name?

Type (string) derived from here
Variable gets value from this node if empty

[View all references](#)

Usage

☐ Topic (limited scope)

☒ Global (any topic can access)

☐ Allow to carry between sess

☐ External sources can set valu

☐ External sources can receive

Variable description ⓘ

Sensitive data ⓘ

☐ Off

Work with Topics

Topics

1

Conversation Start: Greets users and introduces the agent and its capabilities. Triggers when an agent first engages with a user in conversation.

2

Conversational boosting: Creates generative answers from external data sources. Triggers when the agent can't find a match for the user query

3

End of Conversation: Confirms with customers that their query is answered. Triggers from a redirection. Call this topic from your custom topics when you're ready to end a conversation.

4

Escalate: Informs customers if they need to speak with a human. Triggers when "talk to agent" is matched or the Escalate system event is called.

5

Fallback: Informs users their query couldn't be matched to a topic and asks them to try again. Triggers when the agent can't match the user's question or message to a topic.

Topics

6

Multiple Topics Matched: Prompts users to choose their intended topic and sets a system variable to identify the topic triggered. Triggers when a user's message closely matches multiple topics.

7

On Error: Informs customers that an error occurred. Triggers when an error occurs during the conversation.

8

Reset Conversation: Resets the conversation by clearing variable values and forcing the agent to use the latest published agent content. Triggers with a redirection.

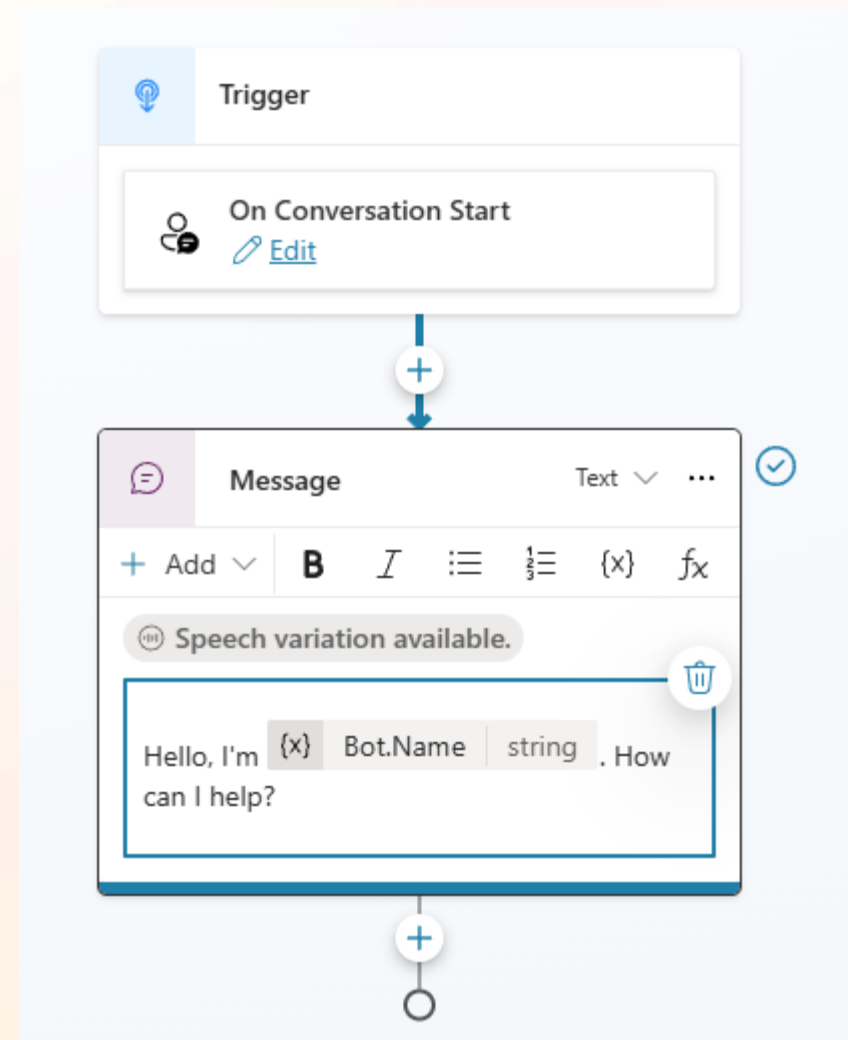
9

Sign in: Prompts customers to sign in when user authentication is enabled. Triggers at the beginning of the conversation when users are required to sign in, or when the conversation reaches a node that uses authentication variables.

Conversation Start topic

First topic run

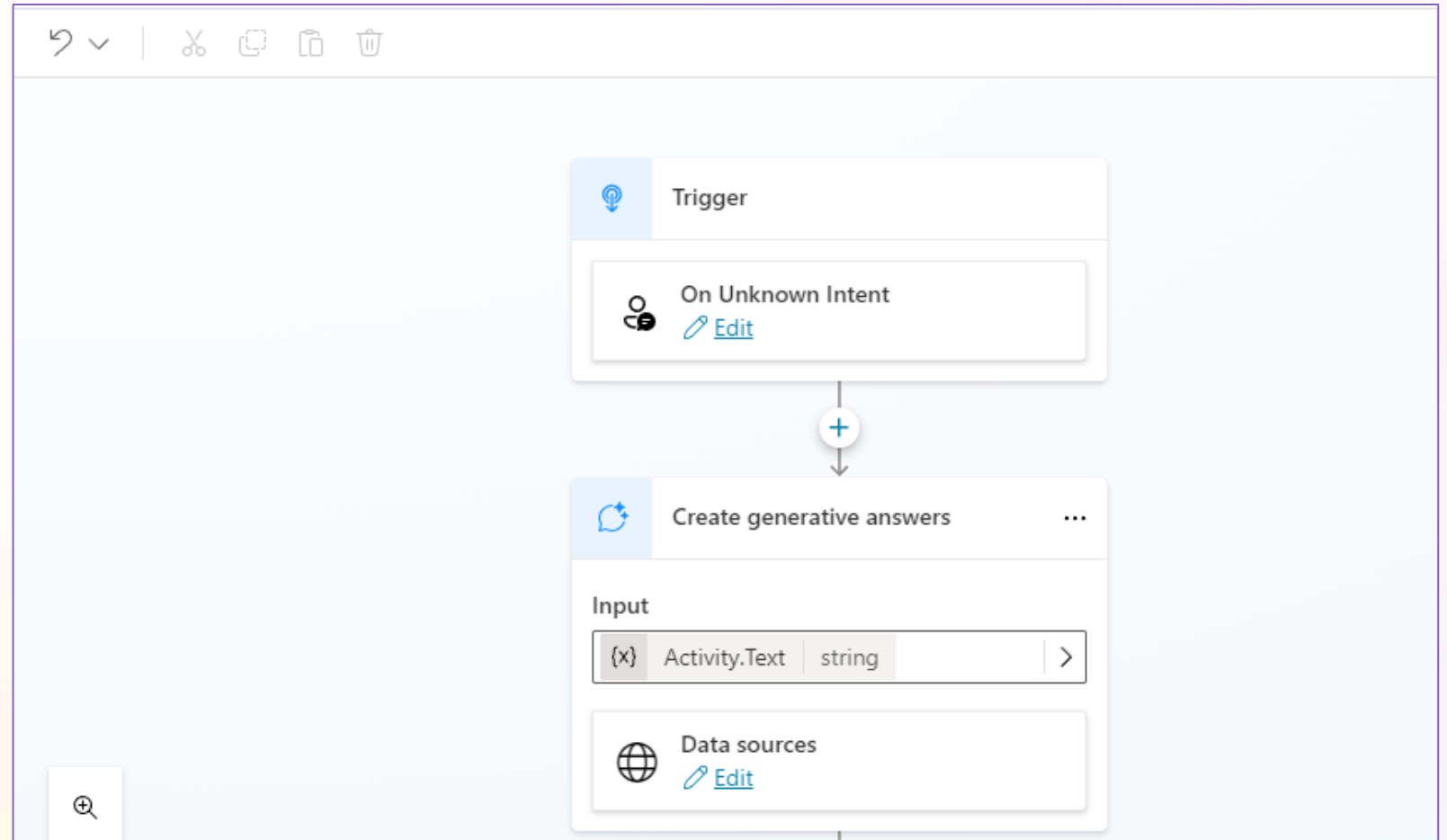
- Contains the first message sent by the agent (the welcome message)
- Can be customized or disabled to change the agent's initial behavior
- Disabling it can reduce cost
- Not the same as the **Greeting** topic



Conversational boosting topic

Generative answers node

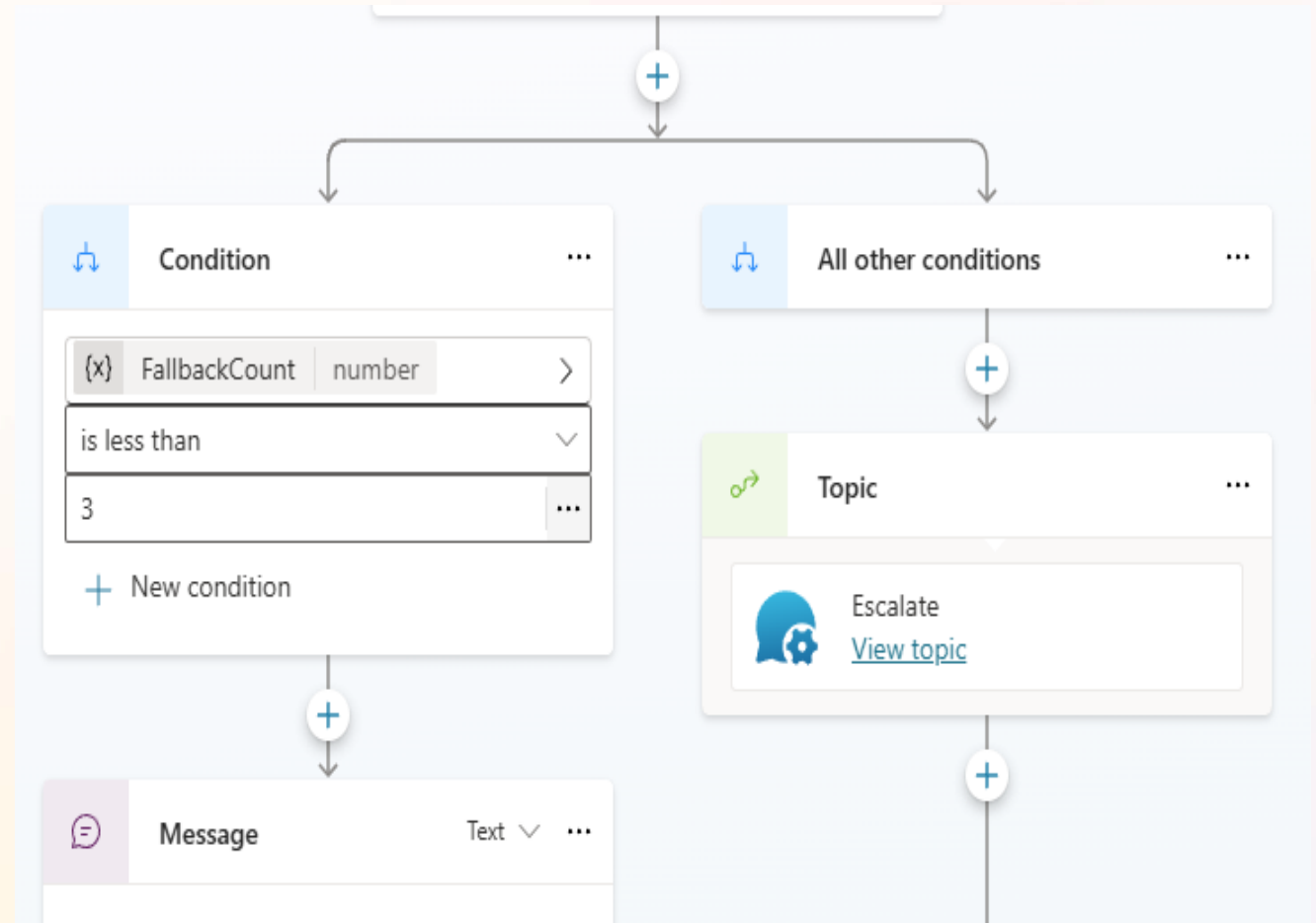
- Add knowledge
- Search only selected sources
- Web search
- Allow the AI to use its own general knowledge
- Content moderation level
- Save LLM response



Fallback topic

On Unknown Intent trigger

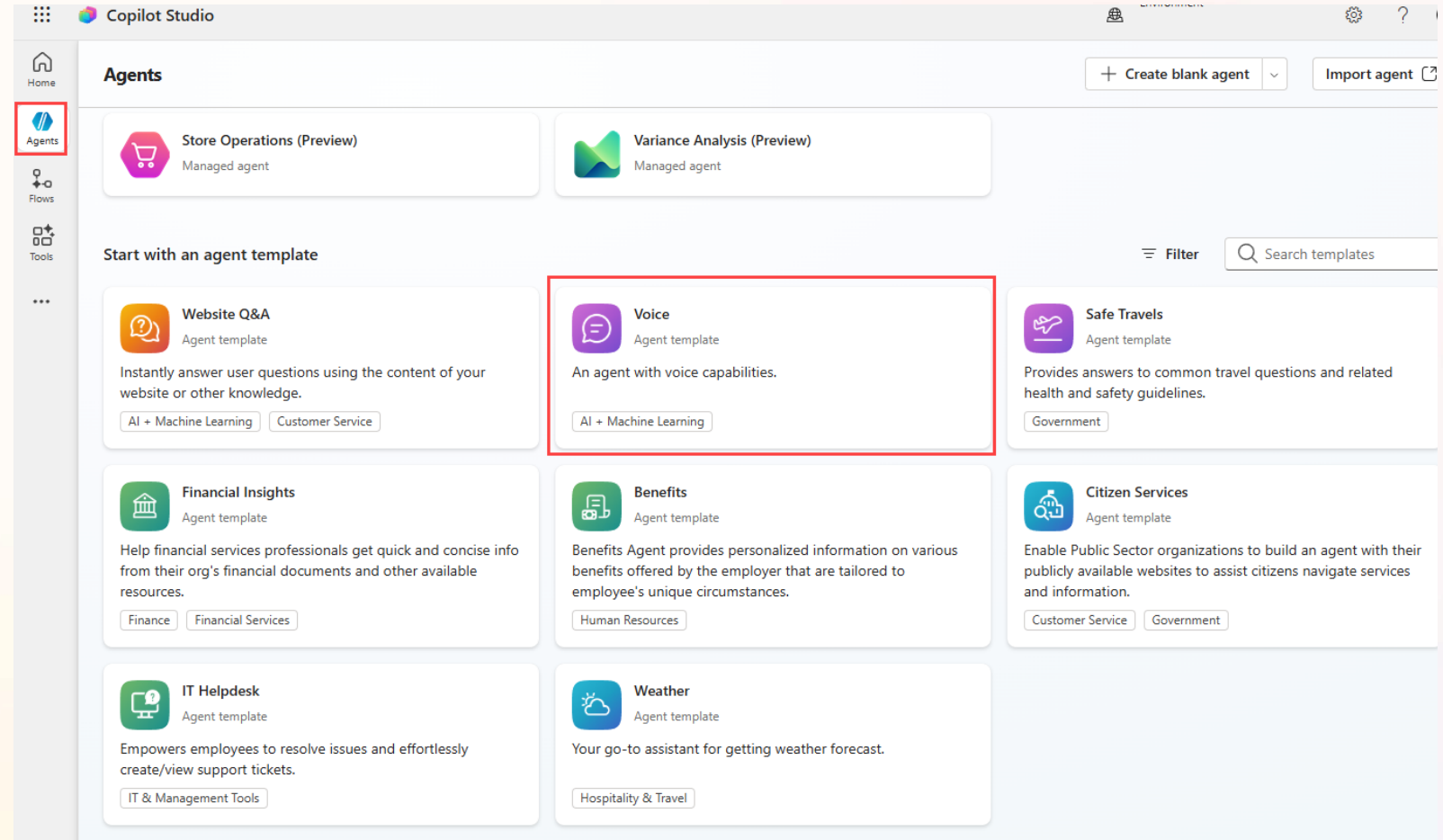
- Called after Conversational boosting topic
- Standard message *I'm sorry, I'm not sure how to help with that. Can you try rephrasing?*
- Redirects to the **Escalate** topic after two times
- Can add a **Create generative answers** node when no escalation is possible



Voice agents

Voice agent template

- Telephony channel turned on by default
- Speech and DTMF are enabled
- Supports interactive voice response (IVR)



Voice-enabled agent topics

- 1 **Main menu** topic

- 2 The **Conversation Start** topic redirects to the Main menu topic

- 3 **Silence detection, Speech unrecognized, Unknown dial pad press, Answering Machine Detection** topics

- 4 The **Conversational boosting** topic is not included

- 5 Sample lesson topics showing how to topics work with Speech and DTMF key tones

Practice Labs

Lab 2: Manage topics in Copilot Studio agents

Module Recap

In this module, we:

Learned how to create topics using Copilot

Learned how to add and configure nodes to create the conversation path

Learned how to work with entities and variables



